Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/08/2012
PI Name:	Straume, Tore Ph.D.		
Project Title:	Lunar EVA Dosimetry: Small Active Dosimetry System for Lunar Extravehicular Activity Missions: Spacesuit and Tool-Box Applications		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **ARS**:Risk of Acute Radiation Syndromes Due to Solar Particle Events (SPEs)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-3943
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Division		
PI Address 2:	MS236-7		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	11/01/2007	**End Date:**	10/31/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: title changed per NSBRI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas Engineering Experiment Station) Lusby, Terry (NASA Ames Research Center)		
Grant/Contract No.:	NCC 9-58-RE01302		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this project is the development of a compact active dosimetry system that can provide real-time radiation monitoring beyond low-Earth orbit (LEO). The dosimetry system would be capable of radiation monitoring during both ambient and solar particle event (SPE) conditions. It is designed to be responsive to the doses, dose rates, and radiation qualities (including secondary neutrons) expected in interplanetary space. This "mini-TEPC (tissue equivalent proportional counter) for deep space" builds on the extensive (and successful) performance history of the much larger tissue-equivalent proportional counters that have been on the Space Shuttle and on the International Space Station (ISS) for two decades, but which are much too large and power intensive to be used beyond LEO. The key innovations of our mini-TEPC are its compact size, low power, and no saturation during large SPEs, while at the same time maintaining the TEPC advantages of tissue equivalence and lineal-energy response. These advances in miniaturization and power are made possible in part by our incorporation of an innovative variance-covariance data collection and processing approach. As astronauts travel beyond LEO and explore asteroids, the Moon, and Mars, there will be a critical need for compact active personal radiation monitoring. Of particular concern is the need for real-time dose-rate measurements during SPEs so that astronauts can seek prompt shelter. SPEs may occur unexpectedly and with highly variable dose rate and can potentially result in high radiation doses to astronauts. The dynamic radiation environment in space requires the development of suitable detection systems that can detect a broad range of dose rates as well as a complex mixture of radiation types. This project is a joint effort between NASA Ames Research Center (ARC), Colorado State University (CSU), and Texas A&M University (TAMU). Importantly, two separate National Space Biomedical Research Institute (NSBRI) grants support this effort (Radiation Effects-RE01301--Principal Investigator Borak and RE01302--Principal Investigator Straume) and reports from both of these grants should be viewed together to appreciate the total effort. CSU designs and fabricates the TEPC sensors, TAMU designs and builds the pre-amplifier with charge integrator, and ARC designs and builds the miniaturized electronics package with bias voltage power supply and microprocessor and programs the microprocessor to calculate dose-mean lineal energy (yD), quality factor, dose, and dose-equivalent. In this project, hardware specifications required by NASA for this technology have either been met or exceeded. We have shown that the tissue equivalent sensor has omni-directional response and can detect radiation with linear energy transfer (LET) between 0.2 and 300 keV/µm. We have also shown that the TEPC monitor can detect maximum credible SPEs without saturation, has good precision and time resolution, consumes only 0.75 W of power, and has a mass of only about 250 g. Hence, it is adaptable to either a real-time personal dosimeter or an area monitor. Additional efforts are, however, required to fine-tune software related to yD calculations, which effect precision of the dose-equivalent rate measurements, but not absorbed dose rate measurements of greatest importance during SPEs. Also, there is a need for additional testing to better define the lower limit of detection, which is important during ambient (non-SPE) conditions. It is recommended that follow-on efforts should be undertaken by NASA to advance this proof-of-principle prototype with the aim of answering the fundamental question: Should this mini-TEPC technology approach be part of NASA's suite of detector systems for deep space missions? Additional evaluations of this technology should be performed to obtain a clear answer to this question.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The compact real-time radiation monitoring capability developed in this project may be useful in homeland security and Department of Defense (DOD) applications, where first responders would require small personal dosimeters capable of detecting real-time mixed gamma and neutron radiations. It may also be applied in high-energy research or medical accelerator facilities where mixed radiations are produced.
Task Progress:	During year 4 (final year for this grant) we developed and fabricated final components, assembled, and tested. The tissue-equivalent sensor was redesigned to improve mechanical reliability. The original design used a single crimp fastener to tension the anode wire and make the electrical connection between the anode and the charge integrator input. This crimp connector could slip in the high molecular weight anode insulator when it was being connected or disconnected from the integrator, resulting in a loose or broken anode wire. This problem was eliminated by replacing the aluminum flange with one machined from stainless steel. The completed detector and charge integrator assembly was tested using both gamma-ray and neutron sources, with data acquisition accomplished using a commercial data logger and later with the miniaturized electronics package. The relative variance minus the relative covariance (v-c) was then calculated. The slope of the change in output voltage was determined and the calibration factor for the detector/charge integrator was determined. The calibration factor was then used to convert v-c, which is calculated in volts, to dose mean single event specific energy in Gy, and the geometry and mass of gas in the detector was used to convert single event specific energy to lineal energy in keV/µm. This approach produced results consistent with published values of the dose mean lineal energy for the radiation sources measured. The final electronic schematics and board fabrication files were developed and prototype fabricated for testing with radiation sources. The design consisted of a main board, daughter board, and sensor interface board. These boards fit into a compact aluminum enclosure. The prototype can be programmed in ANSI C, operate from battery power, communicate through USB through a simple terminal, and connect to a standard evaluation board to measure apparent diffusion coefficient (ADC) performance. Each unit has auxiliary user pins for connection of additional inputs or outputs. C-programming was developed for control and for processing the algorithm based on the data conditioning and results of the Excel analyses. A custom LabView control interface was developed and is capable of graphing dose rate and dose equivalent rate along with providing total integrated dose and dose equivalent. In conclusion, hardware requirements specified by NASA for this technology have either been met or exceeded: tissue equivalent sensor, omni-directional response, LET response between 0.2 and 300 keV/µm, detects maximum credible SPE without saturation, measures dose rate to required level of precision and time resolution, very low power consumption (about 0.75 W), low mass (about 250 g), and adaptable to either a personal dosimeter or area monitor. Additional software work is required to improve yD calculations, and more tests are required to better define lower limit of detection, important for ambient dose-equivalent rates in space.
Bibliography Type:	Description: (Last Updated: 03/22/2019)

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